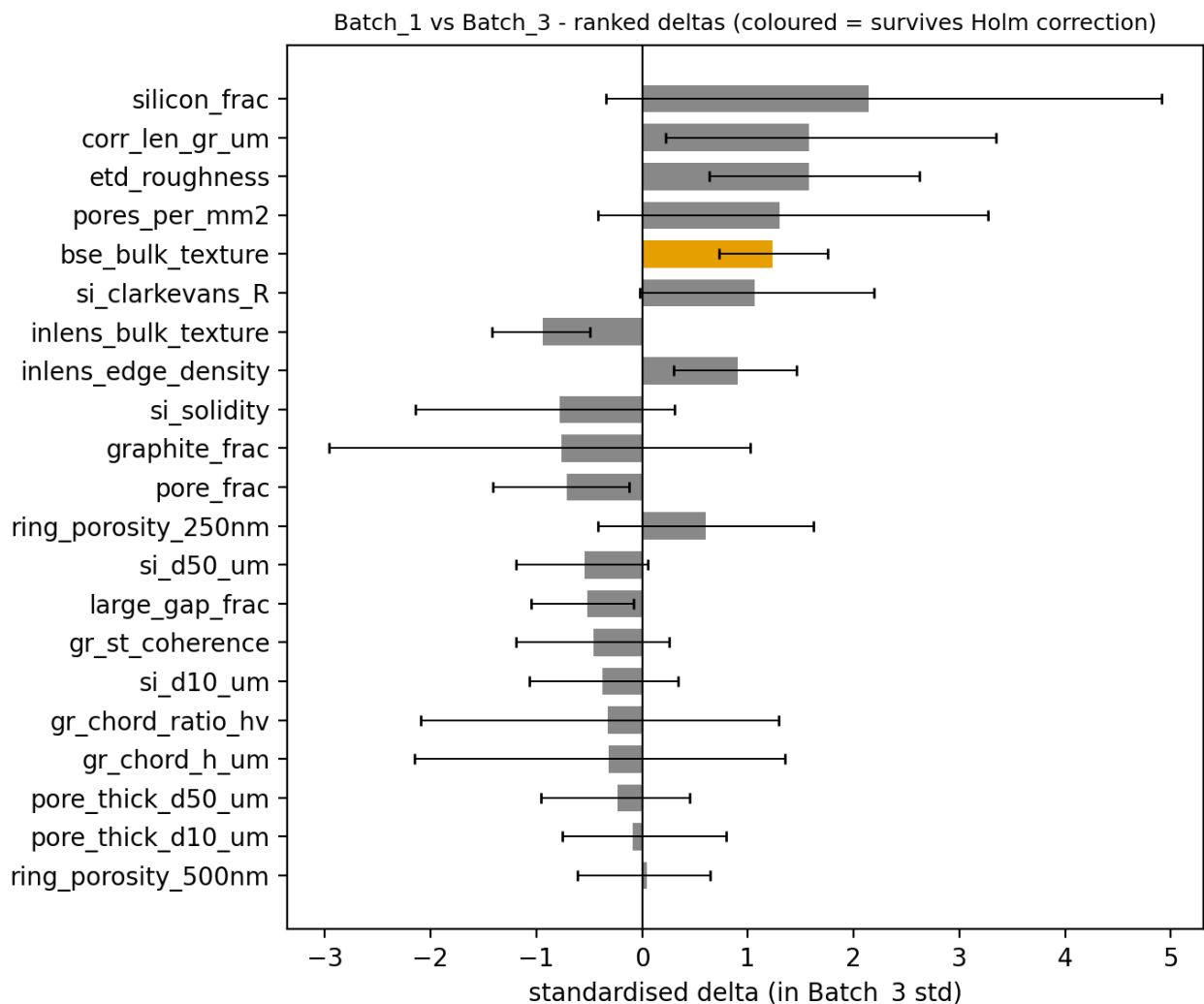


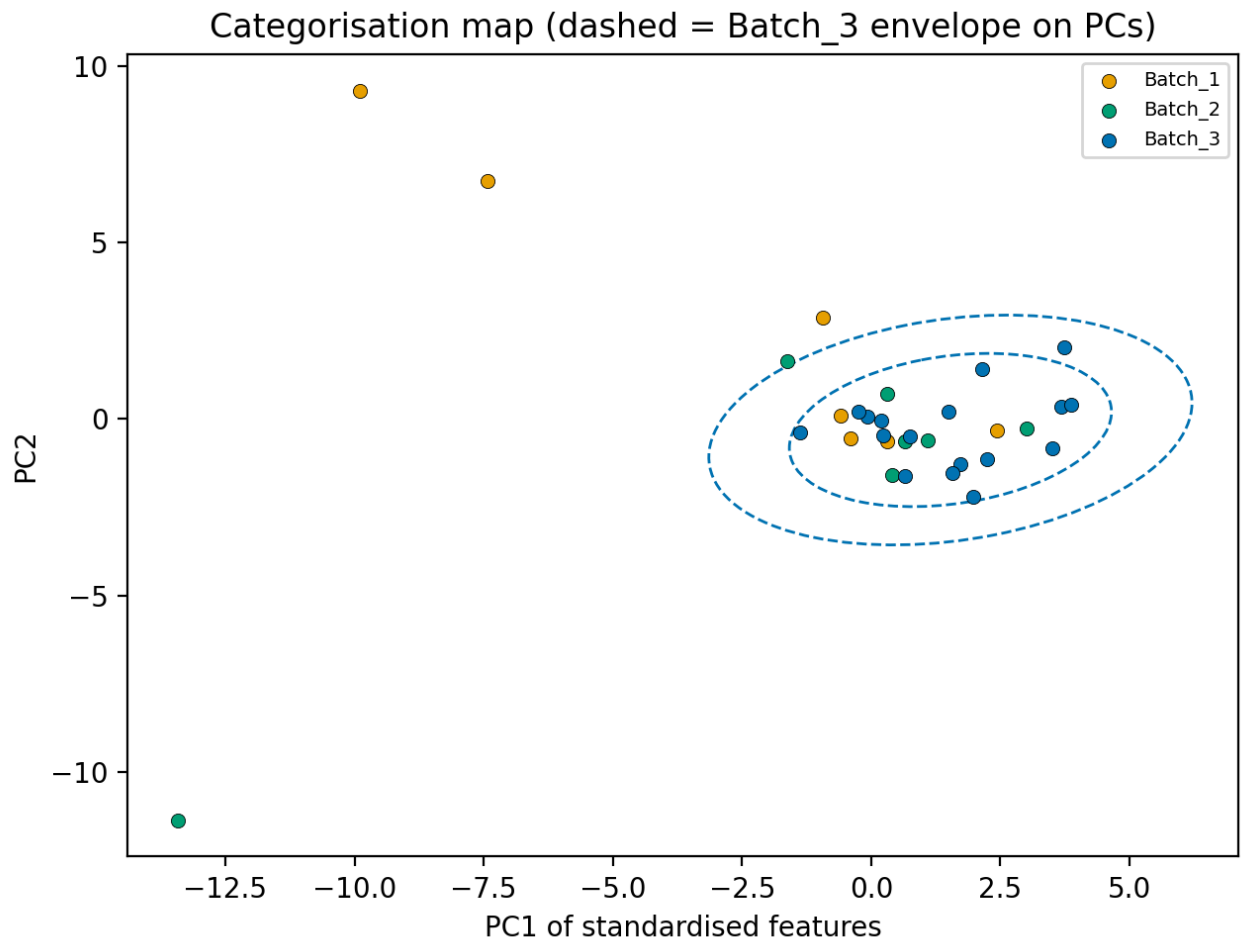
Batch delta analysis - summary

Batch_1 differs from the Batch_3 baseline mainly by having less visible pore space (~13%) and far fewer large gap pores (~2/3 less) while silicon fraction and particle sizes are unchanged - consistent with a denser / harder-pressed electrode. Batch_2 is statistically close to Batch_3 on morphology. Both B1/B2 show a stronger imaging-noise signature than B3, so part of the apparent difference is acquisition, not material. Single images cannot be categorised reliably; a folder of ≥ 5 images can: ~100% for B1-like samples, ~80-100% for B2-like at $n=7$.

Assumption (applies throughout): the grey bulk is assumed graphite, bright particles are assumed silicon-based, black is pore/crack. This comes from image appearance and is unconfirmed.

Baseline: Batch_3 $n=17$ images, all segmented identically. Batch_1: pore_frac -13.6% (suggestive, CI excludes 0 but does not survive Holm), large_gap_frac -71% (suggestive), bse_bulk_texture +18% (different, survives Holm but tracks imaging noise). Batch_2: bse_bulk_texture +15% (different, noise-tracked), gr_st_coherence -8% (suggestive); everything else equivalent or undetermined. Prior quick-pass findings agree in direction (less pore space in B1, B2 close to B3); the +45% silicon signal in the quick pass came from the two low-contrast flagged images and disappears without them.





Reliability: leave-one-image-out which-batch accuracy 42% overall - poor for single images; the categoriser must be used on groups of ≥ 5 images. False-alarm rate on held-out Batch_3 singles: 35%.